

INFO-I 211 - *Information Infrastructure II*

Practice Quiz - Python and Programming

Do not open until told to do so.

Answer questions in the space provided. Draw a box around your final answer if necessary.

Name: _____

IU Email: _____

Which section do you attend?

- ☐ Alexander Hayes - M/W - 9:35–10:50 AM
- ☐ Erika Lee - M/W - 11:10–12:25 AM
- ☐ Alexander Hayes - Tu/Th - 3:55–5:10 PM

General Instructions:

1. You have **60 minutes** to complete this exam. If you must leave this room: you must submit your exam.
2. Using any electronic device (watches, phones, earbuds, etc.) will be met with consequences. Put them away.
3. Include all syntax. e.g. If a program prints a list of strings, its output should look like: `['a', 'b', 'c']`.
4. The **maximum score** is 20 points, including bonus.
5. Questions are arranged roughly by difficulty, with easier ones at the beginning and harder ones at the end. The following table breaks down how you might spend your time:

Question	Lower Question Time	Upper Question Time	Lower Clock Time	Upper Clock Time
Q1	7 min.	10 min.	07:00	10:00
Q2	11 min.	15 min.	18:00	25:00
Q3	14 min.	20 min.	32:00	45:00
Q4	10 min.	15 min.	42:00	60:00
Q5 (Bonus)	18 min.	<i>Skip</i>	60:00	<i>Skip</i>

Question:	1	2	3	4	5	Total
Points:	5	6	4	5	0	20
Bonus Points:	0	0	0	0	3	3
Score:						

1. Answer questions about the following Python program:

```
1  x = 5
2  y = 3
3  print((x * y) + 1)
4  y = 2
5  print((x * y) + 1)
```

(a) [1 pt.] What is the **data type** of each variable?

x:

y:

(b) [1 pt.] Line 3 has an expression inside a print. What values are **x** and **y** bound to when line 3 is evaluated?

x =

y =

(c) [1 pt.] Line 5 has an expression inside a print. What values are **x** and **y** bound to when line 5 is evaluated?

x =

y =

(d) [2 pts.] What final result is shown when this program runs?

2. Answer questions about the following program.

```
1  def foo(x: int) -> int:
2      return x * 2
3
4  def bar(x: int) -> int:
5      if x % 2 == 0:
6          return (x * 3) + 2
7      return (x * 3) + 1
```

(a) [2 pts.] As written, does running this program show/output anything? Why or why not?

(b) [4 pts.] Assume the functions `foo(x)` and `bar(x)` take integer inputs and return integer outputs. Fill in the following table, starting from a function input `x`, and showing what is returned (1 pt. per row).

Input <code>x</code>	<code>foo(x)</code>	<code>bar(x)</code>	<code>bar(foo(x))</code>
0			
1			
2			
3			

3. Consider the following Python program:

```
1  # Initialize variables for the program
2  i: int = 0
3  v: int = 1
4
5  # Compute a result
6  while (i < 4):
7      if (i % 2 == 0):
8          v = v * 2
9          i = i + 1
10
11 # Handle showing outputs
12 while (i != 0):
13     print(i, v, i * v)
14     i = i - 1
15     v = v - 1
```

- (a) [$\frac{1}{2}$ pt.] How many **expressions** are in the print statement on line 13? (*Reminder:* a variable by itself is also an expression. You can also think of this question as asking: “how many things are printed?”)
- (b) [$\frac{1}{2}$ pt.] How many times does the `if` statement on line 7 evaluate to True?
- (c) [1 pt.] What does the variable “v” contain between lines 10-11?
- (d) [2 pts.] What final output gets shown after running this program?

4. [5 pts.] There are several errors in the following program. It is supposed to:

1. Get 6 inputs from the user (using `input()`)
2. Convert them from strings to integers and append them to a list
3. Iterate over the list, printing numbers that are both **divisible by 2** and **less than or equal to 10**

But there are several errors. **Circle and correct 5 errors** (there may be more). Errors may be syntactic or semantic. (*Hint*: Some errors might be easier to identify by tracing the program.)

```
1  # Declare variables
2  A: list[int] = []
3  i: int = 0
4
5  # Read input from the user
6  i = 0
7  while (i < 6):
8      A[i] = int(input("Choose a number> "))
9      i = i + 1
10
11 # Process data and output results
12 while (i < 5):
13     if (A[i] % 2 == 1) or (A[i] < 10):
14         print(A[])
15     else:
16         print(A[i])
17     i = i + 1
```

5. (Bonus) We want to simplify this program:

```

1  v1 = 2
2  v2 = 3
3  v3 = 2
4  v4 = 0
5
6  while (v4 < 4):
7      v1 = v1 + v1
8      v4 = v4 + 1
9
10 while (v3 < v4):
11     print(v1 - v3)
12     v3 = v3 + 1

```

(a) [1 point (bonus)] Which variable(s) are initialized/created, but never used/updated?

(b) [1 point (bonus)] What **values** are in **v4** and **v1** when the first **while** loop (lines 6-8) finishes (e.g. at line 9)? You **do not** need to fill in the table, but it may help track variable state before making your final answer.

Loop iteration	v4	v1
Before the loop		

(c) [1 point (bonus)] An equivalent program can be written which uses only one variable. Complete the following program's **v1** variable (line 1), and **while** loop condition (line 3). (*Hint*: What does the original program print?)

```

1  v1 =
2
3  while (
4      print(v1)
5      v1 = v1 - 1

```